

Arin Balyan

+91-9518108395 | arinbalyan19@gmail.com | arinbalyan.vercel.app | linkedin.com/in/arinbalyan | github.com/arinbalyan | leetcode.com/arinbalyan

Experience

BrilliED / SCG Foundation | AI/Full-Stack Engineer

Nov 2025 – Apr 2026

FastAPI, Python, Vue.js, LangChain, PostgreSQL, Docker, CI/CD, AWS

Remote — Full-time

- Owned development of brillied.in, working directly with the founder to define product features, architecture, and release cycles.
- Implemented LLM optimization techniques (prompt engineering, retrieval refinement, response validation), improving response consistency and reducing failure cases in generated outputs.
- Handled production debugging and system issues, improving reliability across multiple releases.
- Improved system efficiency and request handling, enabling $\sim 3\times$ **traffic growth** while maintaining stable performance under increased load.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face

LLM / Agentic AI: LangChain, LangGraph, RAG, Prompt Engineering

DevOps / Cloud: Docker, AWS, CI/CD, Git, Cloudflare

Backend: FastAPI, Node.js, REST APIs

Databases: PostgreSQL, Redis

Tools: Linux, Kaggle, Streamlit

Projects

EmotionML | Real-Time Emotion Recognition Platform

Jul 2025 – Aug 2025

PyTorch, MobileNetV2, FastAPI, React.js, OpenCV

[GitHub](#)

- Trained **9 deep learning models** on **35,000+ images** (FER-2013 & AffectNet); achieved **98.99% on CK+48** (controlled dataset) and **72.4% on FER-2013** (real-world dataset), outperforming baselines by **12%**.
- Optimised MobileNetV2 reducing model size **60%** (23MB $\rightarrow$ 9MB) and inference time **40%** (250ms $\rightarrow$ 150ms), enabling edge deployment on 512MB RAM devices.
- Developed FastAPI backend for real-time inference, enabling efficient image processing workflows with optimized model integration.

Image Enhancement Web Service | AI Photo Processing Pipeline

Jan 2026 – Present

Python, FastAPI, PyTorch, Real-ESRGAN, GFPGAN, OpenCV

[GitHub](#)

- Engineered production-ready service combining **Real-ESRGAN** (4 $\times$ super-resolution), **GFPGAN** face restoration, and adaptive white-balance/exposure correction.
- Processed high-resolution images with optimized inference pipeline, reducing average processing time and enabling efficient batch workloads.
- Developed an automated image analysis pipeline to intelligently detect and apply required enhancements; supports **batch processing** and **Google Drive** integration via FastAPI.

AgroAid | ML-Powered Agricultural Assistant

Nov 2024 – Feb 2025

Python, TensorFlow, XGBoost, Flask, NLP

[GitHub](#)

- Constructed and optimized an ensemble model (Random Forest + XGBoost + stacking meta-learner) on **50,000+ records** from 3 government datasets achieving **89% prediction accuracy on crop yield dataset**, outperforming single models by **15%**.
- Engineered multilingual recommendation engine in **5 languages**, serving **500+ daily queries** with **sub-200ms response time**.

Education

Bachelor of Technology — Computer Science (AI & ML)

Aug 2022 – May 2026

Vellore Institute of Technology

CGPA: 8.45 / 10

Courses & Certifications

[Applied Machine Learning in Python](#) — University of Michigan

Score: 94%

[MERN Full-Stack Developer Certification](#) — Ethnus

Score: 90%

Achievements & Leadership

- Top 25 Global** among **500+ teams** — Zelestra $\times$ AWS Hackathon; Designed and deployed an ML solution end-to-end in **48 hours**.
- GSSoC Open-Source Contributor:** 30+ merged PRs across 20+ repositories including ML projects and GenAI projects.